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Decoupling Hydroclimatic Controls on Displacement of Retrogressive Thaw Slumps in Qinghai-Tibet Plateau

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Key Points:

- Across 1,034 retrogressive thaw slumps (RTS) in Beiluhe region, lands sink at 12–19 mm/yr with a seasonal amplitude of 27 mm
- Below ~0°C, faster subsidence occurs in drier years; above ~0°C, faster subsidence occurs in wetter years
- Precipitation and land surface temperature jointly modulate RTS surface displacement

Supporting Information:

Supporting Information may be found in the online version of this article.

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Abstract The retrogressive thaw slumps (RTS), a prevalent form of thermokarst hazard in permafrost regions, are increasing in both number and extent under contemporary changing environments. On the Qinghai-Tibet Plateau, RTS threaten transportation infrastructure and disrupt ecosystems. However, the combined influence of hydroclimatic drivers on RTS subsidence remains poorly understood. We therefore examine the relationship between surface displacement and precipitation under varying land surface temperature (LST) conditions in Beiluhe region. Our findings reveal an average seasonal displacement amplitude of 27 mm and a mean annual subsidence of 12–19 mm across RTS. When LST remains below approximately 0°C, faster annual subsidence is linked to drier years. In contrast, when LST exceeds 0°C, faster subsidence is associated with wetter years. These results provide new insights into the coupled dynamics of surface displacement, precipitation, and LST in RTS subsidence, enhancing our understanding of permafrost degradation processes amid climate crisis.

Plain Language Summary Retrogressive thaw slumps (RTS) are a type of mass-wasting process that occurs in permafrost areas when ice-rich ground thaws, causing the land to collapse. They are happening more often and covering larger areas as the climate changes. On the Qinghai-Tibet Plateau, these slumps reshape the land, damage roads, and disturb ecosystems. We still lack knowledge on how precipitation and land surface temperature (LST) work together to affect the subsidence of RTS. In this study, we used satellite radar data to measure how the ground moves in an RTS cluster region and compared our surface movement measurements with precipitation and LST records. We noted that RTS areas typically move seasonally by 27 mm and sink by 12–19 mm every year. Areas that move more seasonally also sink faster overall. When land surface temperatures stay below freezing, faster sinking tends to happen in drier years. But when temperatures rise above freezing, faster sinking happens in wetter years. These findings help explain how precipitation and LST combine to regulate RTS activities, giving us a better understanding of how permafrost systems are degrading nowadays.

1. Introduction

Retrogressive thaw slumps (RTS) are a typical type of thermokarst hazard on Qinghai-Tibet Plateau (Lantz & Kokelj, 2008; Lewkowicz & Way, 2019; Y. Li et al., 2024). It is a type of slope failure in the permafrost region, often initiated by active layer degradation (Huang et al., 2021), water erosion (Y. Yang et al., 2023), or anthropogenic disturbance (Nesterova et al., 2024). Upon the formation of the RTS, ice-rich permafrost in the headwall is exposed to the air, facilitating the melt process and retrogressive collapse of the slumps' headwall (Luo et al., 2022a). Escalating number and expanding area of RTS with easily recognizable land surface alteration are alarming consequences of permafrost degradation during changing climate (Y. Lin et al., 2025).

The detrimental consequences of RTS are manifold. RTS can damage vegetation covers and reshape the permafrost landscape. The associated surface subsidence and topographic changes heighten threats to infrastructure safety, particularly to the roadbeds in Qinghai-Tibet Engineering Corridor (X. Hu et al., 2025; Wu et al., 2024). The evolution of RTS also disturbs the ecosystems (Lantz & Kokelj, 2008; Wu et al., 2024). The release of sediments and water might have cascading impacts on downstream hydrological and ecological systems (Kokelj et al., 2021; D. Li et al., 2021; Mukherjee et al., 2024). This process may weaken the natural capacity for essential services like water regulation and soil conservation, prolong ecosystem recovery, and reduce ecosystem

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resilience (Bartleman et al., 2001; Wei et al., 2023). The evolution of RTS alters the balance of soil carbon, nitrogen, and phosphorus, and the microbial decomposition rates, and it also releases considerable organic carbon from the soil (Nesterova et al., 2024; Xia et al., 2022; Y. Yang et al., 2023). In the meantime, the land also sinks as RTS evolve. Synthetic Aperture Radar (SAR) has become a critical approach to monitor surface displacement due to the wide image swath, regular temporal acquisition, and accurate surface displacement monitoring using Interferometric Synthetic Aperture Radar (InSAR) techniques (Lu & Dzurisin, 2014). For example, B. Hu et al. (2019) obtained InSAR-derived surface displacement rates of slumps along the Qinghai-Tibet Highway and verified the results using field measurements. Z. Jiao et al. (2023) reported faster displacement of 375 RTS using InSAR technique in the northern Qilian Mountains than those in other permafrost areas. InSAR-based studies primarily focus on quantifying the spatial distribution and land displacement of RTS, but the knowledge about seasonal displacement characteristics and controlling factors of RTS remains limited.

It has been widely recognized that changes of precipitation and land surface temperature (LST) modulate the surface processes in permafrost environments. The precipitation increases the moisture content in the active layer, reduces soil shear strength, and promotes the likelihood of continued RTS development (D. Yang et al., 2023). Additionally, an increasing air temperature accelerates permafrost thawing and destabilizes the soil mass (Y. Li et al., 2024; Liu et al., 2015). Current studies mainly rely on field surveys, statistical modeling, and machine learning approaches to investigate the occurrences and evolution of RTS. For example, Balser et al. (2014) used time-series correlation analysis of InSAR and meteorological data to reveal over 80% of RTS in the northwest of Alaska occurred between 2004 and 2005. The timing of these events correlated with unusually early high air temperatures and heavy precipitation during that spring. This study highlighted the critical role of short-term extreme weather conditions in triggering slumps. R. Zhao et al. (2016) observed a time lag between air temperature peaks and maximum subsidence in the Danxiong-Yangbajing area on the southern Qinghai-Tibet Plateau, suggesting a delayed response of frozen ground to air temperature. Lewkowicz and Way (2019) demonstrated that ice-rich continuous permafrost terrain is highly vulnerable to changes in warm summers. Additionally, Yin et al. (2023) used random forest algorithms to assess the environmental drivers of RTS along the Qinghai-Tibet Engineering Corridor, and found that extreme summer climate events (e.g., maximum air temperature and rainfall) contributed the most to the RTS occurrence over the flat areas with fine-grained soils. G. Yang et al. (2025) reported that multiannual variations in RTS on the Qinghai-Tibet Plateau strongly correlated with ground temperature, air temperature, and thawing index, and had a negative correlation with the decrease in downward shortwave radiation. Although the controlling roles of ground temperature, precipitation, and soil moisture in RTS subsidence have been addressed separately, their interactive mechanisms have not been systematically investigated.

To quantify the dynamics of surface displacement and evaluate hydroclimatic controls on permafrost degradation, we focused on RTS, a typical thermokarst hazard. RTS can amplify permafrost-degradation signals relative to the surrounding land surface (Figures S1 and S2 in Supporting Information S1). We analyzed their surface displacements through time using the Copernicus Sentinel-1 SAR images and separated out the secular and periodic time series to reveal the displacement trends and seasonal amplitude. More importantly, we employed a conditional correlation analysis to quantify the relationship between displacement and precipitation under varying LST conditions. These correlations help interpret hydroclimatic influences on RTS subsidence, inform process-based modeling of RTS evolution, and contribute to interpreting permafrost degradation dynamics.

2. Materials and Methods

2.1. Study Area

Beiluhe locates in the northwestern part of the Qinghai-Tibet Engineering Corridor in the central Qinghai-Tibet Plateau (34.56°N–35.45°N, 92.13°E–93.23°E; Figure 1). The region is dominated by ice-rich permafrost, with over 70% of the area having ice content >30% (Luo et al., 2015). Permafrost thickness ranges from 20 to 80 m and the active layer thickness ranges from 1.8 to 3.0 m. The mean annual ground temperature (MAGT) ranges from −2°C to −0.5°C, and annual precipitation total is around 300 mm, with more than 80% precipitation falling between May and August (Z. Lin et al., 2016; Luo et al., 2015). High ice content, shallow active layer, and near-zero ground temperatures make this region highly dynamic in its hydrothermal conditions and a natural laboratory for studying permafrost degradation under changing environments.

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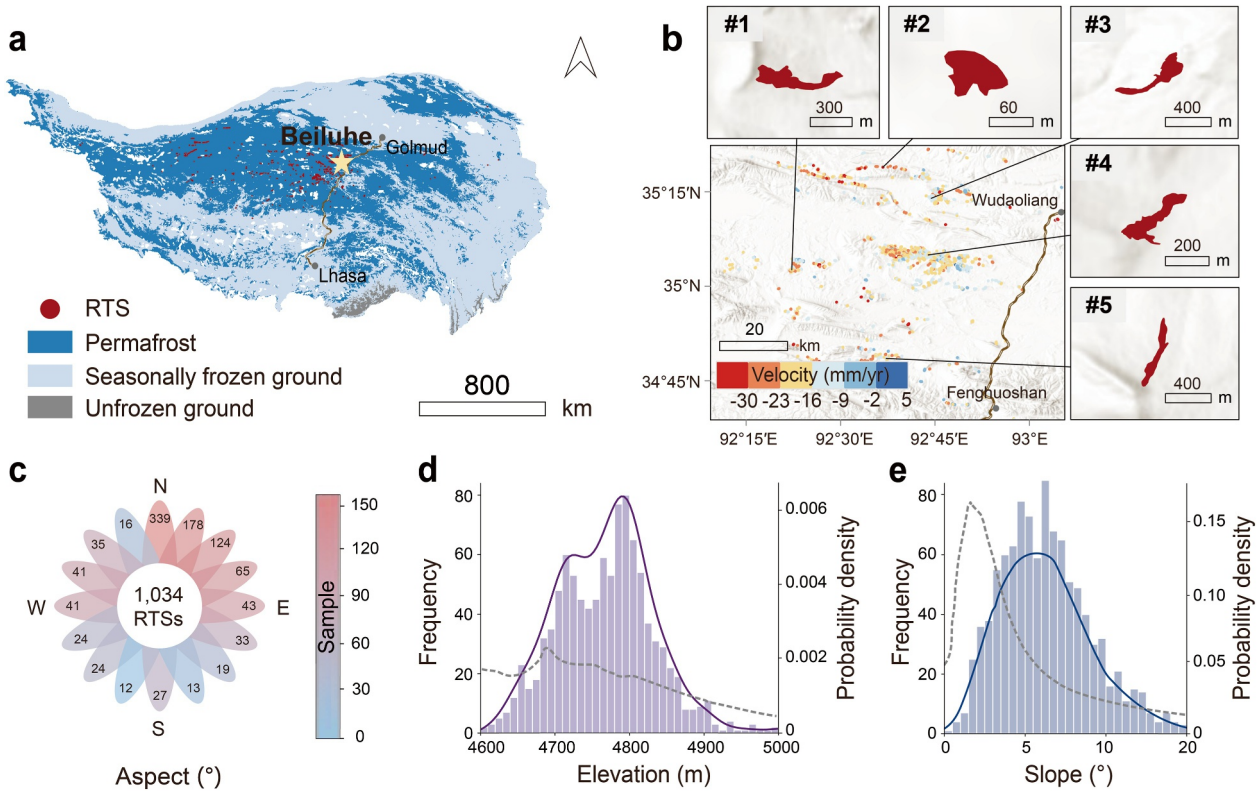

Figure 1. Study area over the Qinghai-Tibet Plateau. (a) The spatial distribution of retrogressive thaw slumps (RTS) in Beiluhe region (locations obtained from Luo et al. (2022a)). Background colors show the geographic locations of permafrost, seasonally frozen ground, and unfrozen ground. (b) Surface displacement velocity in Beiluhe region during 2016–2021; negative values represent subsidence. Five representative RTS sites (#1–#5) were selected to illustrate their morphologic disparities. Their time-series displacement results are shown in Figures S3–S6 in Supporting Information S1. (c–e) Slope aspect, elevation, and slope statistics of 1,028 RTS, respectively. In panels (d, e), solid lines show the probability density of the average topographic variable values at RTS pixels, and dashed lines represent that over all pixels in the study area. Their difference indicates that RTS are not randomly distributed but preferentially occur under specific topographic conditions.

2.2. Data Sets

Luo et al. (2022a) provided the RTS inventory across Qinghai-Tibet Plateau with their geographic locations and geometric boundaries available. A total of 1,034 RTS covering approximately 17.5 km^2 were reported in Beiluhe region. We generated time-series displacement spanning 2015–2022 in a spatial resolution of 30 m using ascending and descending Copernicus Sentinel-1 SAR data. Topographic variables (elevation, slope, and aspect) were derived from the 30-m-resolution Shuttle Radar Topography Mission (SRTM) Digital Elevation Model (DEM). LST data (2015–2022), with a spatial resolution of 1 km and a temporal resolution of twice daily (daytime and nighttime), were obtained from the Thermal and Reanalysis Integrating Moderate-resolution Spatial-Seamless data (Tang et al., 2024). Precipitation data, with a spatial resolution of 0.1° and a temporal resolution of 1 hr, were obtained from the Global Multisource Merging-and-Calibration Precipitation data set (Ma et al., 2025). Figure 2 shows the time-series LST and precipitation averaged across all RTS.

2.3. Time Series Decomposition and Residual Test

The RTS displacement time series (from January 2015 to September 2022) were derived from InSAR data processed using ISCE and MintPy. The line-of-sight displacement was decomposed into vertical direction, and the time series was interpolated on a daily basis. Coherence over RTS was generally high, with 93% of ascending-track pixels and 83% of descending-track pixels exhibiting coherence values greater than 0.7 (Figure S7 in Supporting Information S1). Please refer to Text S1 in Supporting Information S1 for InSAR processing and Text S2 in Supporting Information S1 for uncertainty analysis.

Considering the seasonal freeze-thaw cycles in RTS and the evident annual periodicity in LST, we disentangled the time-series displacement and LST using a 1-year-period sinusoidal function, along with a linear term to

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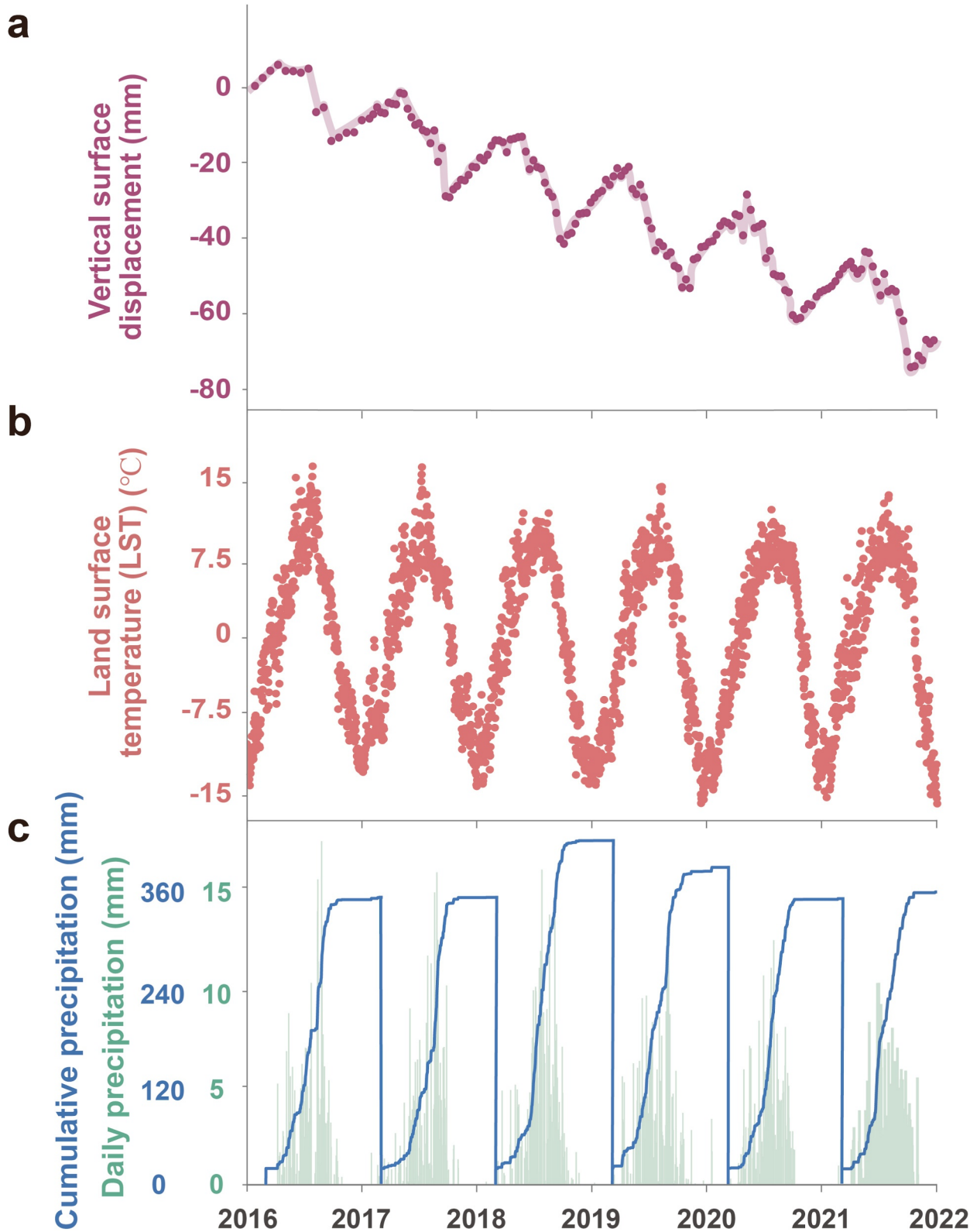

Figure 2. Averaged time series of vertical surface displacement, land surface temperature (LST), and precipitation for all retrogressive thaw slumps in Beiluhe region. (a) Vertical surface displacement derived from two-dimensional decomposition of ascending and descending line-of-sight measurements. Points represent decomposed results, and line denotes the interpolated time series at daily resolution. (b) LST time series. (c) Daily precipitation (green lines) and cumulative precipitation (blue lines, reset to zero in March).

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capture multiannual trends (X. Hu et al., 2021). Assuming a general 1-year period for both displacement and LST, and an undetermined time shift between the two processes, the time window must exceed 1 year to capture a full cycle of variations for both. We tested different lengths of the time window, and the residual analysis suggested the optimal length of 1 year and 8 months. Therefore, for each year, the model fitting time window was set to span from the previous September to the following April. The time series at each RTS was modeled as:

$$y(t) = A \cdot \sin\left(\frac{2\pi}{T} \cdot t + \phi\right) + B \cdot t + C \tag{1}$$

where t denotes the number of days since the start of the fitting period. The sinusoidal function component captures seasonal dynamics with a period T of 365.25 days. The amplitude A characterizes half the seasonal range, and ϕ is the phase shift used to extract seasonal peaks and troughs. The linear term B represents the surface displacement or LST trend, and C denotes the offsets.

To ensure the correctness of time-series decomposition, we applied a two-step residual test. First, the absolute mean residuals were not more than 1 mm for displacement or 0.01°C for LST. Second, residuals were assessed using the autocorrelation function to diagnose remaining temporal dependence (Diodato et al., 2025; Hill et al., 2021; Ray et al., 2021). The decomposition was considered viable when the autocorrelation of residuals falls below 95% confidence with a relatively large time lag (e.g., greater than 15 days and we tested the results until 90 days). In total, 82% of the RTS displacement time series passed these two criteria. Then, we extracted the seasonal amplitude 2A, the annual change ($B \times 365.25$), and the sequential days in one calendar year of the annual peak and trough. The same procedure was applied to the LST time series. These extracted peaks and troughs were further used to compute the time difference between displacement and LST (See Section 4.1 for details). Five representative RTS sites (#1–#5 in Figure 1b) were used to demonstrate the procedure time-series decomposition and residual test (Figures S3–S6 in Supporting Information S1).

2.4. Conditional Correlation Analysis—Kendall's τ_b

To assess the relationship between surface displacement and precipitation under varying LST, we used a conditional correlation analysis. The rank-based, non-parametric association measures are commonly used to assess the direction and strength of the relationship between two random variables. Among the most widely applied in this category are Somers' D, Goodman-Kruskal's γ, and Kendall's τ (Metsämuuronen, 2021). Such methods do not assume any specific data distribution and are well suited for detecting monotonic relationships. To quantify the relationship between annual surface displacement and annual precipitation under varying LST, we employed a conditional correlation analysis based on Kendall's τ_b. Kendall's τ, originally proposed by Kendall (1938), comes in two variants: τ_b and τ_c. In this study, we selected τ_b for two main reasons. First, τ_b performs reliably even when some observations share identical values for a variable. For example, if two samples both have the same precipitation value but different displacement rate, the original Kendall's τ may yield a slightly biased correlation between the two variables since it does not adjust for such repetition. But τ_b can address this issue by modifying the denominator to exclude the number of repeated-value samples. This ensures that the resulting correlation estimate remains accurate (Kendall, 1945). Second, τ_b has demonstrated greater robustness and statistical efficiency when the sample size is small (Arndt et al., 1999). Kendall's τ_b is computed as:

$$\tau_b = \frac{n_c - n_d}{\sqrt{(n_0 - n_X)(n_0 - n_Y)}} \tag{2}$$

where X denotes annual surface displacement and Y denotes annual precipitation. In this study, subsidence is recorded as negative displacement; for the correlation analysis we therefore used the absolute displacement $|X|$, so that larger values indicate faster subsidence. Each RTS sample is a paired record $(|X|, Y)$. For example, (20, 300) means one RTS sample with annual surface displacement of −20 mm/yr and annual precipitation of 300 mm. Given n RTS samples, the total number of distinct pairwise comparisons is $n_0 = n(n-1)/2$. For any two samples $A = (|X_a|, Y_a)$ and $B = (|X_b|, Y_b)$, the pair is concordant if $(|X_a| - |X_b|)(Y_a - Y_b) > 0$ (both variables move in the same direction), and discordant if $(|X_a| - |X_b|)(Y_a - Y_b) < 0$ (one increases while the other decreases). For illustration, $A = (20, 280)$ and $B = (25, 300)$ are concordant; by contrast, $A = (20, 280)$ and $C = (15, 350)$ are discordant. Here, n_c and n_d are the numbers of concordant and discordant pairs. A tie in displacement occurs when two samples have

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the same magnitude ($|X_a| = |X_b|$) but differ in precipitation ($Y_a \neq Y_b$); a tie in precipitation occurs when $|X_a| \neq |X_b|$ but $Y_a = Y_b$. Then n_X and n_Y are the numbers of tied pairs in displacement and precipitation, respectively, with $n_X = \sum_i t_i(t_i - 1)/2$ and $n_Y = \sum_j u_j(u_j - 1)/2$, where t_i and u_j are the numbers of samples in the ith $|X|$ tie group and the jth Y tie group. Here i indexes the distinct $|X|$ values and j indexes the distinct Y values. Additionally, pairs tied on either variable are excluded from n_c and n_d. Overall, a larger absolute value of Kendall's τ_b indicates that the association between the two variables is more evident. The positive Kendall's τ_b indicates that faster surface displacement is associated with higher precipitation; conversely, the negative value indicates that faster surface displacement rate is associated with lower precipitation.

The surface displacement is highly sensitive to both thermal and hydrological conditions, we conducted a binned correlation analysis to investigate the relationship between annual surface displacement and annual precipitation under different LST conditions. Each RTS was treated as one sample and assigned to a bin based on the annual mean LST at a given year. For the LST and precipitation data rasters, we overlaid each RTS polygon on the grid. If a polygon was entirely contained within a single pixel, we assigned that pixel's value to the RTS. If a polygon intersected multiple pixels, we assigned their mean value. Quantile divisions were then applied with bin counts of 10. Within each bin, Kendall's τ_b was calculated between annual displacement and annual precipitation. This conditional binning approach allowed us to examine how the strength and trend of the correlation change with varying LST.

3. Results

3.1. Time Series of Displacement and LST

The spatial average of vertical displacement across all RTS has been interpolated to generate continuous displacement time series (Figure 2a). The cumulative vertical displacement exhibits clear seasonal amplitude superimposed on a secular subsiding trend. The seasonal amplitude was relatively modest in 2016 and 2017 but intensified in subsequent years. From 2018 onward, the seasonal cycles became more distinct, with pronounced subsidence during thaw seasons. Overall, the ground surface underwent a cumulative vertical subsidence of about 80 mm, with an annual subsidence of 12–19 mm/yr (negative displacement means subsidence).

Seasonal characteristics were extracted from displacement and LST time series that passed the residual test. The seasonal displacement amplitude is 27 mm on average. Seasonal displacement typically peaks in early July to mid-August, and troughs appear from late winter to early spring. LST reaches the highest in late July and the lowest in mid-January, and the seasonal amplitude is 22.49°C.

Seasonal displacement amplitude at RTS is positively correlated with surface displacement rate ($r = 0.4371$, $p < 0.0001$). This means that RTS with greater seasonal movements spanning freezing and thawing seasons also sink more abruptly, likely due to aggravating permafrost degradation and thickening the active layer.

3.2. Topographic Characteristics of RTS

Slope aspects reveal a regular directional pattern in RTS occurrences. About 57% of RTS occur on north- to northeast-facing slopes (Figure 1c), where solar radiation is relatively limited. This results in lower MAGT and prolonged snow covers during wintertime; both favor ground-ice preservation. When spring arrives, snowmelt infiltrates into near-surface soils and often leads to active layer saturation, reducing the shear strength and destabilizing the slopes. These radiation-driven disparities in ground thermal conditions play a critical role in the spatial distribution of thermokarst landforms.

About 96% of RTS locate between 4,600 and 4,900 m in elevation, with the highest concentration occurring in the 4,700–4,800 m range (571 RTS, 55%) (Figure 1d). This mid-altitude zone is characterized by widespread ice-rich permafrost and relatively shallower active layers. At higher altitudes, thin weathered layers and low ice content limit the formation of RTS, while at lower elevations, the scarcity of ice-rich soils and the presence of extremely gentle slopes appear to hinder their initiation and growth (Luo et al., 2022a).

About 85% of RTS occur on gentle slopes between 2° and 10° (Figure 1e). Flat terrains may promote ice formation but lack sufficient gravitational force to initiate slumping. In contrast, steep slopes tend to drain quickly and prevent ice accumulation. Therefore, gentle slopes represent a favoring environment for RTS where the moisture retention and gravitational stress are ideal.

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These topographic patterns are robust to DEM choice, with consistent aspect, elevation, and slope distributions across the SRTM DEM, the Advanced Land Observing Satellite (ALOS) Global DEM, and the Copernicus DEM (Figure S8 in Supporting Information S1).

4. Discussion

4.1. Time Difference Between Surface Displacement and LST

Seasonal surface displacement extrema over RTS show a clear time difference compared to the LST extrema. Our time-series decomposition was conducted independently for displacement and LST on 1-year and 8-month time windows (see Section 2.3). The displacement troughs follow LST peaks with a longer delay than the lag between displacement peaks and LST troughs. This asymmetry may suggest the variant rates of heat conduction during active layer freeze-thaw processes. During summer, subsidence is delayed due to the time required for thawing until sufficient latent heat is attained, compounded by the inherently gradual nature of thermal conduction (Douglas et al., 2020; Lewis et al., 2012; Nicolsky & Romanovsky, 2018). In contrast, frost heave during wintertime occurs more rapidly because cold ground can transfer heat efficiently; therefore, the frost heave reacts faster (Throop et al., 2012).

The time difference between LST peaks and displacement troughs typically exceeds 90 days, with delays reaching 131 and 126 days in 2016 and 2017, longer than those in other years. Table S1 in Supporting Information S1 shows the time difference for each year. The time difference between LST troughs and displacement peaks is generally shorter but follows a similar multiannual variation, with slightly longer time difference in 2016–2017 and shorter time difference after 2018. Our results support a delayed response of displacement to LST, but the time length varies across years due to climate variability.

From 2016 to 2018, both LST peaks and troughs occurred slightly earlier, consistent with advancing displacement extrema (Figure 3). From 2019 to 2021, LST troughs postponed, accompanying with a pronounced delay in displacement peaks. A similar delay can be identified between LST peaks and displacement troughs over this same time. Exceptionally, both LST peaks and troughs postponed in 2018–2019, while displacement peaks and troughs arrived earlier. This indicates possible influences of non-thermal factors, such as the precipitation anomaly. Overall, the first-order temporal variations between LST and displacement are mostly consistent.

4.2. Hydroclimatic Coupling Mechanisms of RTS Subsidence

The quantile bin diagram shows a joint control of surface displacement by precipitation and LST (Figure 4a). Conditional correlation analysis demonstrates that the relationship between surface displacement and precipitation varied substantially with changing LST. RTS have been grouped into bins based on the annual mean LST. A weak negative correlation is observed between subsidence and precipitation in colder conditions (below 0°C, blue box in Figure 4a); subsidence rates increase as decreasing precipitation. However, as LST increases, this association gradually changes into a weak positive correlation. In warmer conditions (above 0°C, pink box in Figure 4a), subsidence rates increase as increasing precipitation. This transition suggests a thermal threshold around the freezing point, beyond which the influence of precipitation on displacement reverses. Repeating the analysis with two alternative precipitation products yields consistent results, supporting the robustness of this transition (Figure S9 in Supporting Information S1).

To further elucidate this correlation reversal, we examined specific temperature bins with the strongest negative and positive correlation (Figures 4b and 4c). In the below 0°C temperature bin (−0.76°C to −0.56°C), subsidence rates decline with wetter climate (increasing precipitation). In contrast, within the above 0°C temperature bin (0.64°C to 2.09°C), subsidence rates increase with wetter climate. In cold conditions, subsurface ice is stable and precipitation falls mainly as snow, limiting the processes of infiltration and subsidence (Machguth et al., 2016; Mohammed et al., 2019; Widhalm et al., 2025). In warmer conditions, increasing precipitation contributes to thaw-induced subsidence by raising soil moisture and facilitating heat transfer within the active layer (O'Connor et al., 2020; Widhalm et al., 2025; Y. Zhao et al., 2022). Our analysis decoupled the interplay roles of precipitation and LST in RTS sinking.

Land surface temperature is widely recognized as the primary control on RTS, because freeze-thaw cycles in permafrost soils are driven by temperature variations, promoting the slump development (Y. Li et al., 2024). Concomitantly, precipitation, in the form of both rainfall and snow, modifies the thermal and stress conditions of

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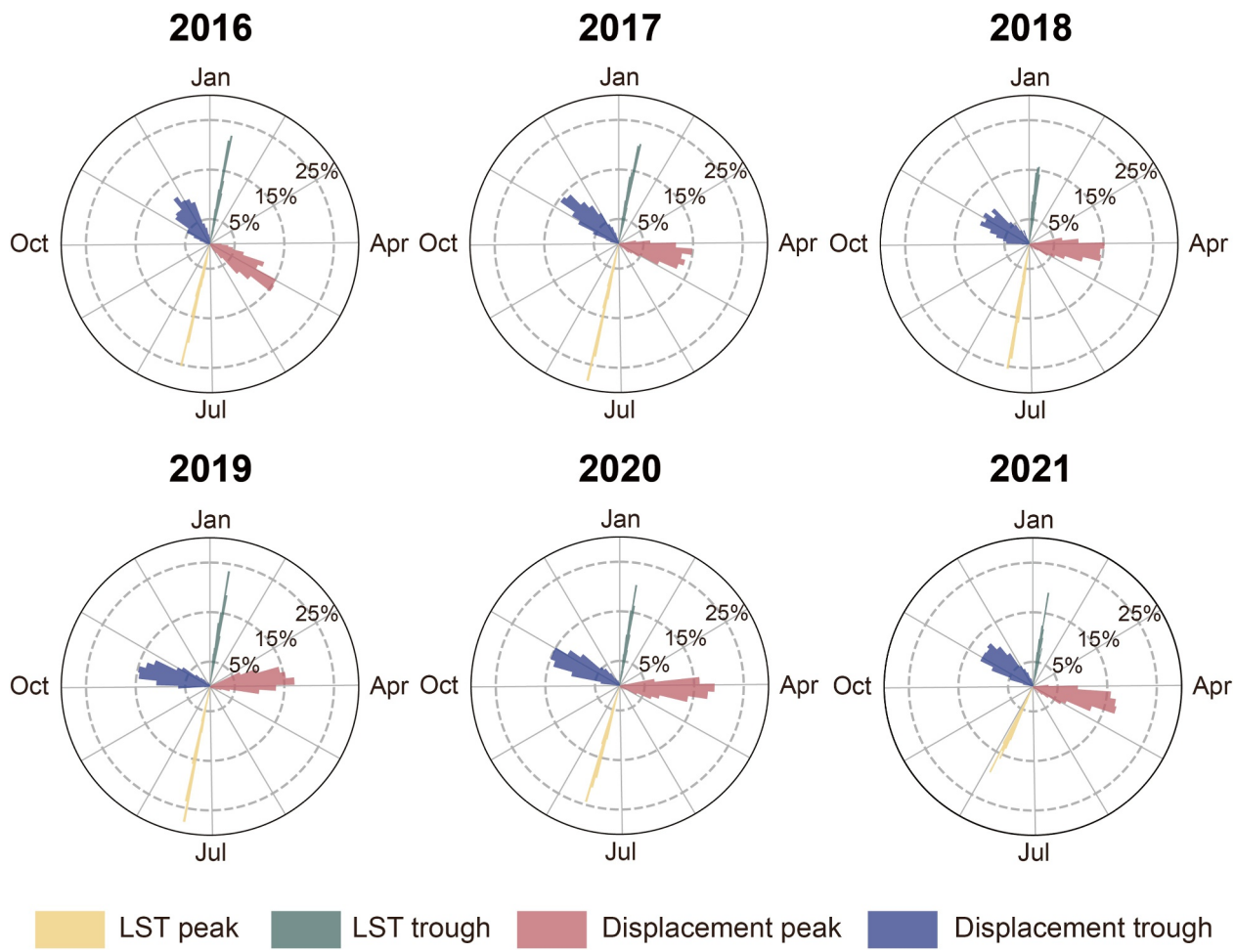

Figure 3. Annual distributions of seasonal peak and trough dates for surface displacement and land surface temperature from 2016 to 2021. The percentage indicates the proportion among all retrogressive thaw slumps.

RTS slopes. Intense rainfall or rapid snowmelt can instantly saturate the active layer, elevate pore-water pressure, reduce shear strength and trigger failures (Y. Li et al., 2024). Our conditional correlation analysis reveals that these two forcings do not act independently but are coupled in regulating the slumps' displacement. The results can provide clues for representing hydroclimatic driving mechanisms in process models of RTS evolution.

In cold conditions, a large fraction of precipitation falls as snow or becomes frozen within the surface layer. Additional precipitation therefore can build up a thick, long-lasting snowpack and increases near-surface ice content, rather than sustaining liquid infiltration to depth (Iwata et al., 2011; Scherler et al., 2010). Although a portion of the water can still infiltrate as liquid, under subzero LST it is predominantly involved in phase-change processes, that is, it refreezes near the surface and forms ice-rich layers (Fu et al., 2022). The associated latent heat release keeps soil temperatures close to 0°C and increases the latent-heat limit to allow for the melting of deeper ground ice (Groenke et al., 2023; Lyu et al., 2024; Yi et al., 2019). Consequently, a larger fraction of the seasonal heat is consumed by shallow freezing-thawing processes, leaving less heat available for deep thaw. Therefore, increased precipitation tends to suppress active-layer deepening and RTS sinking when LST remains below 0°C.

In warm conditions, most precipitation falls as rain onto a more thawed active layer. Infiltrating rainwater can advect heat downward and potentially deepen thaw depth (Devoie et al., 2021; Uhlemann et al., 2021; Y. Zhao et al., 2022). Higher pore-water pressure and reduced shear strength in saturated, ice-rich sediments can thus promote headwall retreat and land subsidence (C. Jiao et al., 2025; D. Yang et al., 2023). Therefore, the precipitation enhances RTS subsidence in this warm regime.

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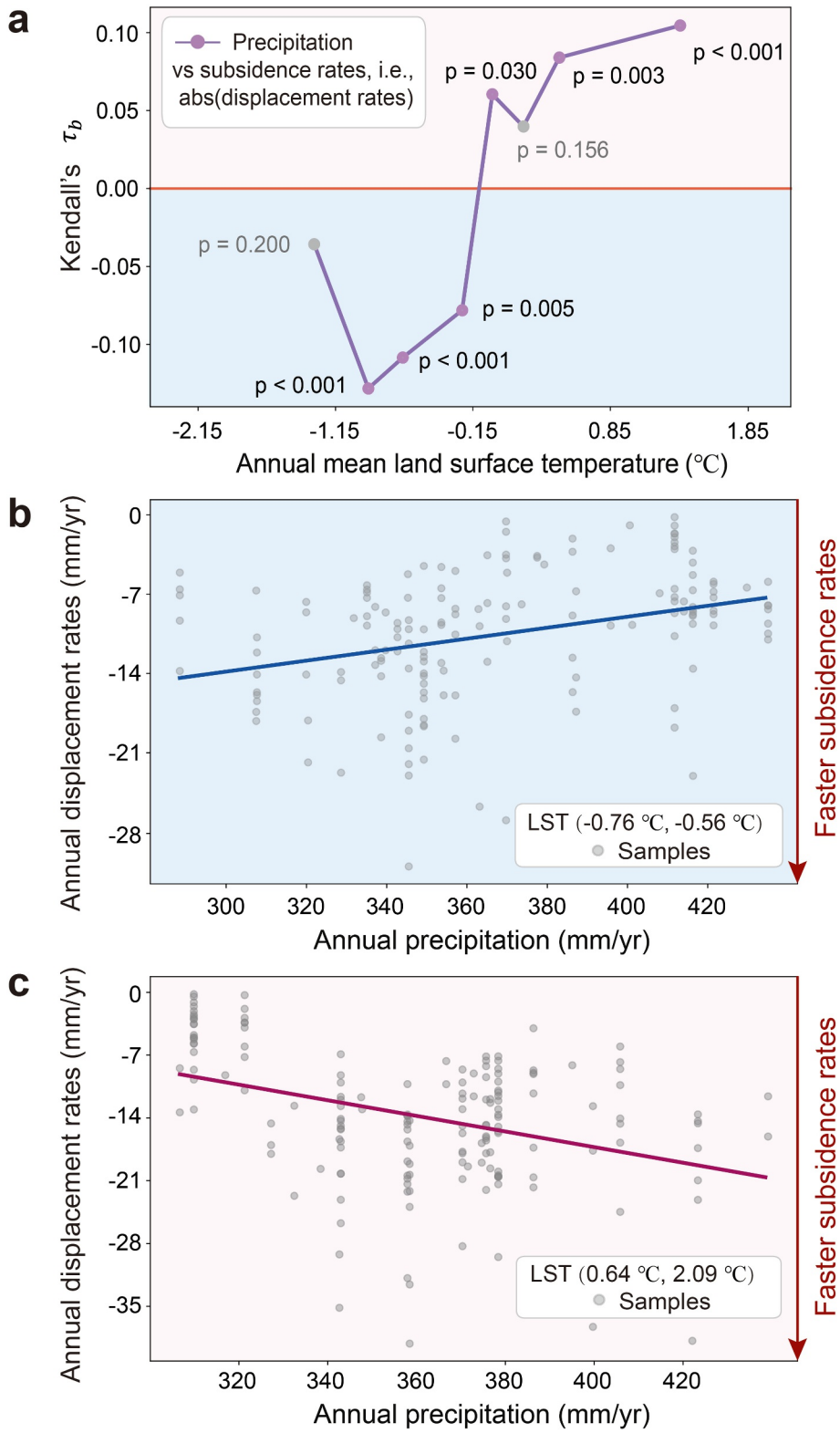

Figure 4.

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4.3. Limitations

A limitation of this study is the absence of external validation data, such as GPS leveling or ground-based thaw depth measurements, to directly assess the accuracy of the InSAR-derived displacement results. But obtaining such data in the RTS areas is challenging. Therefore, we collected published estimates of surface displacement rates from Beiluhe and nearby permafrost areas (see in Table S3 in Supporting Information S1), and our results are consistent with these previous estimates. In addition, uncertainty may propagate from the line-of-sight displacement to the derived vertical component.

We computed the average displacement within external boundaries of each individual RTS, instead of separating out the headwall and slump-floor, due to limited spatial resolution. It does not mean that the RTS is moving uniformly across morphological zones from the headwall to the toe. To describe the spatial variability within each RTS, we calculated the standard deviation across all pixels within each RTS polygon at each temporal acquisition (Figures S10 and S11 in Supporting Information S1).

Temporally, different interpolation methods may result in variant seasonal and annual signals. However, our sensitivity test indicates this impact is negligible: cubic-spline versus linear interpolation changes seasonal amplitudes by ≤0.7 mm and annual subsidence rates by ≤0.4 mm/yr (Figure S12 and Table S4 in Supporting Information S1).

Spatially, another limitation is the resolution of LST and precipitation data sets. The InSAR results have a resolution of 30 m, but hydroclimatic data sets have a much coarser resolution (i.e., LST at 1 km, precipitation at ∼10 km; Figure S13 in Supporting Information S1). This resolution mismatch may dilute local relationships between RTS motion and climate factors, especially for small and unevenly distributed RTS. Despite this limitation, our study area spans multiple LST and precipitation grid cells, and the regional gradients in LST and precipitation provide sufficient spatial disparities for statistical analysis.

5. Conclusions

Focusing on a RTS hotspot in Beiluhe region on Qinghai-Tibet Plateau, we jointly analyzed the seasonal surface displacement amplitude and timing as well as displacement rates, DEM, hydroclimatic variables of precipitation and LST to decouple the hydroclimate forcings in RTS sinking. RTS generally cluster on north- and northeast-facing gentle slopes (2°–10°) in altitudes between 4,600 and 4,900 m. The surface displacement of RTS exhibits pronounced seasonal oscillations in pace with frost heave in winter and thaw subsidence in summer. The land sinks on a multiannual timescale due to an overall thawing trend. RTS exhibiting larger seasonal displacement amplitudes also experience greater long-term subsidence. The time series of surface displacement and LST are out of phase, still with a changing time difference between their peaks and troughs through the years. The relationship between surface displacement and precipitation is modulated by thermal conditions. When the LST is below approximately 0°C, faster subsidence is associated with drier conditions. As the LST increases and surpasses this threshold, faster subsidence instead corresponds to wetter conditions. Our findings clarify the hydroclimatic influences on RTS dynamics and help advance process-based modeling of permafrost degradation, strengthening our understanding of degradation conditions and their evolution.

Figure 4. Effects of land surface temperature (LST) on the relationship between annual surface displacement (negative values suggest subsidence) and annual precipitation over retrogressive thaw slumps (RTS) samples. (a) Kendall's τ_b correlation coefficients (*y*-axis) between surface displacement and precipitation, computed under varying LST conditions (*x*-axis). Each point represents one bin in annual mean LST. *P*-values are annotated, and lower *p* indicates more reliable solutions. Bins with $p > 0.20$ are excluded. Among the remaining bins, those with relatively larger *p*-values are shown in gray to distinguish them from the other bins. (b) Displacement-precipitation relationship in relatively colder LST conditions (−0.76°C to −0.56°C). (c) Displacement-precipitation relationship in relatively warmer LST conditions (0.64°C to 2.09°C). Each point in panels (b, c) corresponds to an RTS sample's subsidence and precipitation statistics in each year during 2016–2021. Solid lines show linear fits of scatter points. Details of the statistical tests associated with these linear fits are provided in Table S2 in Supporting Information S1.

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Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Data Availability Statement

The retrogressive thaw slumps (RTS) inventory is available from Luo et al. (2022b) via https://zenodo.org/records/7309867. The Shuttle Radar Topography Mission (SRTM) Digital Elevation Model (DEM) is available from the U.S. Geological Survey (USGS), Earth Resources Observation and Science (EROS) Center (2019) via https://doi.org/10.5066/F7PR7TFT. Land surface temperature (LST) data during 2015–2022 are available from J. Zhou et al. (2021) via https://doi.org/10.11888/Meteoro.tpdc.271252. Precipitation data during 2015–2022 are available from Ma (2025) via https://doi.org/10.11888/Atmos.tpdc.301878. The boundaries of permafrost categories (permafrost, seasonally frozen ground, and unfrozen ground) are available from L. Zhao (2019) via https://doi.org/10.11888/Geocry.tpdc.270468. The data and codes are available from Y. Zhou et al. (2026) via https://doi.org/10.5281/zenodo.18812010.

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